

Problem

An industrial heater has a nameplate that reads: 210 V 60 Hz 12 kVA 0.78 pf lagging
Determine:

(a) the apparent and the complex power

(b) the impedance of the heater

Step-by-step solution

Step 1 of 2

(A) Apparent power = $S = 12 \text{ KVA}$

$$P = S \cos \theta = (12)(0.78) = 9.36 \text{ KW}$$

$$Q = S \sin \theta = 12 \sin(\cos^{-1}(0.78))$$

$$= 7.51 \text{ KVAR}$$

$$S = P + jQ = 9.36 + j7.51 \text{ KVA} .$$

Step 2 of 2

$$(B) \quad S = \frac{|V|^2}{Z} \Rightarrow Z^* = \frac{|V|^2}{S}$$

$$= \frac{(210)^2}{(9.36 + j7.51) \times 10^3}$$

$$= 2.866 - j2.3$$

$$Z = 2.866 + j2.3 \, \Omega .$$